AESS Sustainability Hackathon 2026

Track 4: Sustainable Space Systems and Orbital Lifecycle

Team Name	Nova Foundry
Project Title	AstroForge - Autonomous Space Debris Capture and Recycling System
Submission Type	Phase 1 - Concept Document

1. Problem Statement

Space debris is a critical threat to orbital infrastructure, with over 34,000 objects (>10 cm) traveling at speeds exceeding 7 km/s. Current solutions primarily involve costly avoidance maneuvers rather than active remediation. The risk of Kessler Syndrome—a cascading collision effect—is increasing annually, potentially rendering specific orbits unusable for future generations.

Key Issue: Existing systems lack a circular economy model. Debris is treated as a hazard to be avoided or de-orbited, ignoring the potential value of materials already in space.

2. Proposed Solution

AstroForge is a 795 kg autonomous spacecraft designed to capture, neutralize, and recycle metallic space debris. Optimized for Sun-Synchronous Orbits (SSO) at 600-800 km, AstroForge implements In-Situ Resource Utilization (ISRU) to convert orbital hazards into usable plasma propellant.

System Architecture:

Subsystem	Function & Implementation
Iris Capture	A 6-blade aperture mechanism with 45°/s servo-controlled closure for debris interception.
Eddy Current Brake	715-turn electromagnetic coil (560W) to decelerate debris without mechanical impact.
Mechanical Damping	Viscoelastic Metamaterial shock absorbers to neutralize residual impact vibrations.
Thermal Recycling	Resistive heating chamber (2000°C) for phase-change conversion of aluminum fragments.
Power Management	1kW Solar array + 5kWh Battery + 20kJ Supercapacitor for high-pulse braking discharge.

3. Implementation & Results

The system was validated using a high-fidelity digital twin in MATLAB/Simulink R2021b. The simulation utilized an 80-second mission profile with 15 kg aluminum debris traveling at 50 m/s relative velocity.

Key Performance Metrics:

Parameter	Baseline (Collision)	AstroForge Performance
Debris Velocity	50 m/s (Fatal impact)	0 m/s in 6.0 seconds (Success)
Braking Efficiency	N/A	76% (Post-thermal dissipation loss)
Energy Consumption	N/A	7 kJ (from 20 kJ supercapacitor)
Attitude Accuracy	N/A	±0.5° (ADCS PID: Kp=200, Kd=350)

4. Impact & Sustainability

AstroForge advances the Debris-as-a-Service (DaaS) model, providing two major sustainability benefits:

• Environmental: Direct removal of high-risk fragments from densely populated orbits, preventing the generation of thousands of micro-debris fragments.

• Economic: Establishing an In-Space Manufacturing (ISM) value chain by selling recycled plasma propellant to orbit-transfer vehicles, significantly lowering fuel launch costs.

5. Limitations & Future Work

• Current Limitation: The simulation assumes high-conductivity metallic debris (Aluminum).

• Next Steps: Implementation of Computer Vision for multi-material classification (Steel/Titanium) and development of a 1:10 scale ground prototype for microgravity verification.

6. Repository & Reproducibility

All simulation files, CAD renders, and technical logs are available in the public repository.
GitHub: https://github.com/Qusaiamer12/AstroForge

7. Conclusion:

AstroForge demonstrates a **technically feasible, energy-efficient, and scalable** approach to active debris removal through **in-situ resource utilization**. The integrated simulation proves:

☑ **Safe debris neutralization** (50 m/s → 0 in 6s)
☑ **Power sustainability** (solar-recharged operations)
☑ **Thermal safety** (coil temp <150°C limit)

This project advances space sustainability by converting orbital hazards into useful resources.

References:

1. ESA Space Debris Office, *Annual Report 2024*
2. Schaub & Junkins, *Analytical Mechanics of Space Systems* (4th ed.)
3. NASA ODPO, *Debris Mitigation Guidelines*
4. Eddy Current Braking: Simuni et al., *J. Spacecraft & Rockets* 2019